

Accessing Linux Data Partitions from Windows

In a dual boot Windows-Linux computer, it is comparatively easier to view the Windows partitions from Linux, than the other way round, i.e., viewing Linux partitions from Windows. In this article, the author presents a way around this challenge.

Today, in the world of computers, dual booting or running both Windows and Linux on the same system, is fairly common. In which case, users often want to transfer files/data between the two operating systems. Linux has native support for Windows file systems (NTFS and FAT), which means that you can access Windows partitions from Linux. But, this is not the case with Windows. To access Linux partitions from Windows, you need third-party software, specifically designed for this purpose.

Basic differences between Windows and Linux file systems

The file systems used in Windows and Linux are completely different in many ways – from how the disks and devices are enumerated to the level of data security provided by them. The file systems used in Windows are FAT (FAT12, FAT16 and FAT32) and NTFS – this last variant is the most used file system in all new Windows operating systems. Linux supports a greater number of file systems like Ext2, Ext3, Ext4, XFS, JFS, etc. The predominant file system used in Linux is Ext3.

On Windows, a disk is divided into partitions and exposed to the user through a drive letter. Even the CD-ROM and removable media get a drive letter. Linux doesn't use drive letters. On Linux, everything is under the root directory (/). Linux treats all the devices as files. All devices like disks and CD-ROMs connected to Linux will be mounted to the root directory or to any other directory under the root like /dev/sda and /dev/cdrom, respectively.

The Windows file system is not case sensitive but Linux file systems are. For example, you can't have file names like *mickey.txt* and *Mickey.txt* in the same directory in Windows. In Linux however, file names like *mickey.txt* and *Mickey.txt* can reside in the same directory.

On Windows, the application locks exclusive access to files, whereas Linux doesn't lock exclusive access to the file as often as Windows does. For example, after watching a video file in VLC on Windows, if you try to delete it without closing the VLC player or opening another file in it—you'll see an error message. You need to stop watching the file in VLC before you can delete it, rename it, or do anything else to it. But Linux allows you to delete the file once it is closed from the VLC player.

Tools used to access the Linux partition from Windows

There are various open source tools available on the Web that you can use to access the Linux partitions from a Windows system. These are categorised based on their uses and functionality.

- Tools used to read and write on the Linux partition
- Tools used for 'read only' access to the Linux partition

Tools use to read and write on the Linux partition

Ext2Fsd*: This tool provides options to assign drive letters, to flush the cache to disk, to view advanced properties for a drive or partition, etc.

